

Stat 286: Causal Inference with Applications, Fall 2025

Section 1: Potential Outcomes*Author: Benedikt Koch***1 Causality**

Intuitively, we know why association is not causation. If someone claims that schools giving tablets perform better, we can point out that those schools are typically wealthier and would likely perform better *even without* tablets.¹ Let's illustrate this with an example we will use throughout the section.

1.1 Example: Tablets in Schools

Let units (here, schools) be indexed by $i = 1, \dots, n$. Define the binary treatment

$$T_i \in \{0, 1\}, \quad T_i = \begin{cases} 1 & \text{if school } i \text{ gives tablets in class,} \\ 0 & \text{otherwise.} \end{cases}$$

Let $Y_i \in \mathbb{R}$ denote the observed outcome (e.g., the average test score of school i). For each school we define two *potential outcomes*:

$$Y_i(0) \text{ (score if not given tablets),} \quad Y_i(1) \text{ (score if given tablets).}$$

The central challenge: we never observe the *pair* $(Y_i(0), Y_i(1))$ for the same unit (school) at the same time. Indeed, if the treatment (tablets) is given to school i ($T_i = 1$), we observe $Y_i(1)$; if no tablets are given ($T_i = 0$), we observe $Y_i(0)$. We write this as the observed outcome equals the potential outcome under the received treatment: $Y_i = Y_i(T_i)$. Thus, we observe one potential outcome per unit, not both.

Suppose (purely for illustration) we could observe *both* potential outcomes for four schools:

i	$Y_i(0)$	$Y_i(1)$	T_i	Y_i	$Y_i(1) - Y_i(0)$
1	500	450	0	500	-50
2	600	600	0	600	0
3	800	600	1	600	-200
4	700	750	1	750	50

The (finite-sample) average treatment effect (ATE) is then

$$\text{ATE} = \frac{1}{4} \sum_{i=1}^4 (Y_i(1) - Y_i(0)) = \frac{-50 + 0 - 200 + 50}{4} = -50,$$

which indicates that tablets would *lower* students' test scores on average.

¹Running example adapted from Matheus Facure, *Causal Inference for the Brave and True*, "Introduction to Causality": <https://matheusfacure.github.io/python-causality-handbook/01-Introduction-To-Causality.html>.

2 The Fundamental Problem of Causal Inference

However, as we eluded to before, for each unit we only observe one of $(Y_i(1), Y_i(0))$, never both. In reality the same data would look like:

i	$Y_i(0)$	$Y_i(1)$	T_i	Y_i
1	500	—	0	500
2	600	—	0	600
3	—	600	1	600
4	—	750	1	750

A naive difference in means is

$$\frac{1}{2} \sum_{i:T_i=1} Y_i - \frac{1}{2} \sum_{i:T_i=0} Y_i = \frac{600 + 750}{2} - \frac{500 + 600}{2} = 125,$$

which *disagrees* with the causal ATE = -50 above. A naïve comparison of means would suggest tablets improve performance and should be rolled out widely; however, the causal effect is actually negative. The gap reflects selection bias (treated and control schools differ systematically) and underscores why careful causal design and assumptions are essential in public policy, medicine, business and many more. Mathematically speaking, the phenomena appears because

$$\frac{1}{4} \sum_{i=1}^n (Y_i(1) - Y_i(0)) \neq \frac{1}{2} \sum_{i:T_i=1} Y_i - \frac{1}{2} \sum_{i:T_i=0} Y_i$$

or in super-population terms (we will come back to that),

$$\mathbb{E}[Y(1) - Y(0)] \neq \mathbb{E}[Y | T = 1] - \mathbb{E}[Y | T = 0],$$

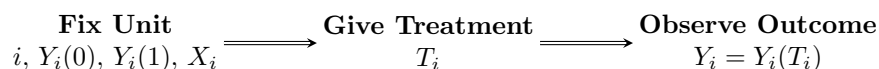
i.e., the causal quantity does not correspond to the statistical one. The main goal of causal inference is to create conditions (or impose assumptions) under which association *becomes* causation by making the groups comparable in their counterfactual outcomes.

3 Potential Outcomes

Causal studies ask *what-if* questions: how would Y change if we would set T to a different value? To formalize this, we use *potential outcomes*. For each unit i and treatment level t , $Y_i(t)$ denotes the outcome that *would* be observed for unit i if, at the time of measurement, the treatment were set to t . Note that t must not need to be T_i and can be different to the observed treatment.

3.1 Unit-Level Determinism

We adopt the Neyman-Rubin framework [Neyman, 1923, Rubin, 1974] of potential outcomes. Under this view, for a fixed unit i the pair of potential outcomes (say for a binary treatment) $(Y_i(0), Y_i(1))$ is a *fixed*, non-random attribute of that unit. This is called *unit-level determinism*. This means, when we fix the unit i , we also fix all the causal path the unit *could* take under the treatment. However, at the time of treatment, we only ever observe one of the potential causal path, $Y_i = Y_i(T_i)$. The causal ordering is illustrated below:



Here X_i denotes pre-treatment covariates (age, race, height, etc.) observed before treatment assignment. Thus potential outcomes are pre-treatment characteristics and not changed by the treatment (but the outcome is).

3.2 Other Schools of Thought

Very briefly, let me point out that there are other, perspectives on potential outcomes. For example, Greenland [1987], VanderWeele and Robins [2012] discuss *stochastic potential outcomes*, where $Y_i(t)$ is random even for a fixed unit i (an approach often used in epidemiology). Others, most prominently Dawid [2000], argue that potential outcomes should be avoided altogether (or used only sparingly), since they can never be observed jointly and thus assumptions about them cannot be tested directly. Instead he argues for a *decision-theoretic* approach to causal inference. Finally, the *structural causal model* (SCM) tradition [Pearl, 2009] replaces potential outcomes with structural equations and DAGs, using the *do*-operator to formalize interventions. This is a very popular framework, especially in computer science. In practice, these frameworks are often complementary, and when assumptions are aligned they typically agree on the same causal quantities.

4 Principal Strata

In the binary case $T_i, Y_i \in \{0, 1\}$ it is often useful to classify units by their *principal stratum*: the potential-outcome pair $(Y_i(0), Y_i(1))$. In the running example, let $Y_i = 1$ indicate that school i 's students meet a proficiency threshold, and $Y_i = 0$ indicate that they did not.

$(Y_i(0), Y_i(1))$	Stratum name	Interpretation
$(0, 0)$	Never-survivors	Fail regardless of tablets
$(0, 1)$	Responders	Pass only if given tablets
$(1, 0)$	Harmed	Pass only if <i>not</i> given tablets
$(1, 1)$	Always-survivors	Pass with or without tablets

These strata are *principal* because membership is defined by potential outcomes and is therefore invariant to assignment. At the unit level, strata are unobserved since we never see both $Y_i(0)$ and $Y_i(1)$.

Unit-level identifiability. From a single unit's observation (T_i, Y_i) we cannot determine its principal stratum:

$$\begin{aligned}
 T_i = 1, Y_i = 1 &\Rightarrow \text{Responder or Always-survivor,} \\
 T_i = 1, Y_i = 0 &\Rightarrow \text{Harmed or Never-survivor,} \\
 T_i = 0, Y_i = 1 &\Rightarrow \text{Always-survivor or Harmed,} \\
 T_i = 0, Y_i = 0 &\Rightarrow \text{Never-survivor or Responder.}
 \end{aligned}$$

Thus the individual effect $\tau_i = Y_i(1) - Y_i(0)$ is not identifiable; we target *average* effects (e.g., ATE) under assumptions.

5 Identification²

5.1 Connecting the counterfactual to the observed world

We distinguish between:

1. *Counterfactual* distribution: $(Y_i(1), Y_i(0), T_i, X_i) \sim \mathbb{P}^*$.
2. *Observational* distribution: $(Y_i, T_i, X_i) \sim \mathbb{P}$.

Here, X_i denotes observed covariates (extra information we observe about the unit *before* the treatment is given). Causal quantities (e.g., ATE) are functions of $\mathbb{P}^*$, but our data are sampled from $\mathbb{P}$. Thus, we can learn about $\mathbb{P}^*$ only through $\mathbb{P}$.

²Motivated by Prof. Matt Blackwell's lecture slides

DEFINITION 1 (IDENTIFICATION) A causal quantity τ is identified if there exists a known map g such that

$$\tau = g(\mathbb{P}),$$

i.e., τ can be written purely as a function of the observational distribution.

In general, linking $\mathbb{P}^*$ to $\mathbb{P}$ requires *assumptions* and often we talk about identification strategies like which assumptions allow you to express the causal estimand as a function of observables?

5.2 Assumptions

What assumptions let us link the counterfactual world $\mathbb{P}^*$ to the observed world $\mathbb{P}$? Unless otherwise stated, we will assume:

- **Causal ordering.** $T_i \rightarrow Y_i$ (no reverse causality/simultaneity).
- **Stable Unit Treatment Value Assumption (SUTVA).**
 - *No interference:* $Y_i(T_1, \dots, T_N) = Y_i(T_i)$; treatments on other units do not affect unit i .
 - *Consistency:* $Y_i = Y_i(T_i)$; the observed outcome equals the corresponding potential outcome.

Often also assumed. Additional identification assumptions (often needed in observational studies)

- **Randomization / Unconfoundedness**
 - *Unconditional (randomized experiments):* $(Y_i(1), Y_i(0)) \perp\!\!\!\perp T_i$.
 - *Conditional (observational studies):* $(Y_i(1), Y_i(0)) \perp\!\!\!\perp T_i \mid X_i$ for a rich set of covariates X_i .
- **Positivity / Overlap.** $0 < \mathbb{P}(T_i = 1 \mid X_i) < 1$ almost surely (each covariate pattern can, in principle, receive either treatment).

5.3 When Assumptions Fail

Tablets in Schools

- *Causal ordering:* may fail if higher-scoring schools choose to adopt tablets *because* of prior performance (reverse causality) or if adoption and outcomes are jointly driven by a contemporaneous grant (simultaneity).
- *No interference:* may fail if district policies, teacher sharing, or competition create spillovers across schools (one school's adoption affects others' outcomes).
- *Consistency:* may fail if “tablet use” varies meaningfully (1:1 vs. shared devices, different apps, teacher training), so $T_i = 1$ does not map to a single well-defined intervention.
- *Unconfoundedness:* more plausible only after adjusting for confounders like wealth, baseline scores, teacher quality, and district resources in X_i . It *fails* if unobserved factors (e.g., leadership quality, parental involvement) affect both adoption and outcomes even after conditioning. If we only compare schools with and without tablets, richer schools may both adopt tablets and have higher test scores, leading us to overestimate the effect of tablets.
- *Overlap:* fails if some strata never/always adopt (e.g., elite private schools always $T = 1$, severely underfunded schools always $T = 0$), yielding no treated-control overlap for those X values.

Flu Shot Scenario: Effect of a flu shot (T) on days sick (Y).

- *Causal ordering:* People vaccinate *after* feeling ill $\Rightarrow Y \rightarrow T$.
- *No interference:* Herd immunity; Y_i depends on others' T_j .
- *Consistency:* “Flu shot” has different versions $\Rightarrow T=1$ not well-defined.

- *Unconfoundedness*: Health-consciousness or access affects both T and Y (selection bias).
- *Overlap*: Some strata always/never vaccinate; no comparable counterfactuals.

EXAMPLE 1 (ATE) Assume (i) SUTVA/consistency $Y_i = Y_i(T_i)$, (ii) ignorability $(Y(1), Y(0)) \perp\!\!\!\perp T$, and (iii) positivity $0 < \Pr(T = 1) < 1$. Then the ATE can be identified

$$\begin{aligned}
 \mathbb{E}[Y(1) - Y(0)] &= \mathbb{E}[Y(1)] - \mathbb{E}[Y(0)] && \text{(linearity of expectation)} \\
 &= \mathbb{E}[Y(1) \mid T = 1] - \mathbb{E}[Y(0) \mid T = 0] && \text{(ignorability \& positivity)} \\
 &= \mathbb{E}[Y(T) \mid T = 1] - \mathbb{E}[Y(T) \mid T = 0] && \text{(by conditioning)} \\
 &= \mathbb{E}[Y \mid T = 1] - \mathbb{E}[Y \mid T = 0] && \text{(consistency).}
 \end{aligned}$$

Note. The conditionals $\mathbb{E}[Y(t) \mid T = t]$ and $\mathbb{E}[Y \mid T = t]$ are well-defined only if $\Pr(T = t) > 0$ for $t \in \{0, 1\}$, otherwise we condition on an event of probability zero.

EXAMPLE 2 (NON-IDENTIFIED: VARIANCE OF THE INDIVIDUAL EFFECT) Let $\tau_i = Y_i(1) - Y_i(0)$. Even under SUTVA, ignorability, and positivity, the variance of the individual treatment effect is not identified:

$$\begin{aligned}
 \text{Var}(Y(1) - Y(0)) &= \text{Var}(Y(1)) + \text{Var}(Y(0)) - 2\text{Cov}(Y(1), Y(0)) \\
 &= \mathbb{E}[Y(1)^2] + \mathbb{E}[Y(0)^2] - 2\mathbb{E}[Y(1)Y(0)] - (\mathbb{E}[Y(1)] - \mathbb{E}[Y(0)])^2.
 \end{aligned}$$

Randomization identifies the marginals $(\mathbb{E}[Y(t)], \text{Var}(Y(t)))$, but not the cross-term $\mathbb{E}[Y(1)Y(0)]$ (equivalently $\text{Cov}(Y(1), Y(0))$), since $Y(1)$ and $Y(0)$ are never observed jointly. Thus $\text{Var}(Y(1) - Y(0))$ is not a function of $\mathbb{P}$ without additional assumptions; at best one can report bounds.

6 Estimation

Identification tells us *what* to estimate, not *how*. Once an estimand is identified, we estimate the corresponding function of $\mathbb{P}$ from data. This is a purely statistical question once identification is established. For instance, a difference-in-means estimator for an identified ATE is

$$\hat{\tau}_{\text{ATE}} \equiv \frac{1}{n_1} \sum_{i: T_i=1} Y_i - \frac{1}{n_0} \sum_{i: T_i=0} Y_i,$$

where $n_t = \sum_i \mathbf{1}\{T_i = t\}$ for $t \in \{0, 1\}$.

Principle. Identification comes first; estimation comes second. Without identification, properties of an estimator are irrelevant.

References

- A Philip Dawid. Causal inference without counterfactuals. *Journal of the American statistical Association*, 95(450):407–424, 2000.
- Sander Greenland. Interpretation and choice of effect measures in epidemiologic analyses. *American journal of epidemiology*, 125(5):761–768, 1987.
- Jerzy Neyman. On the application of probability theory to agricultural experiments. essay on principles. *Ann. Agricultural Sciences*, pages 1–51, 1923.
- Judea Pearl. *Causality*. Cambridge university press, 2009.

Donald B Rubin. Estimating causal effects of treatments in randomized and nonrandomized studies. *Journal of educational Psychology*, 66(5):688, 1974.

Tyler J VanderWeele and James M Robins. Stochastic counterfactuals and stochastic sufficient causes. *Statistica Sinica*, 22(1):379, 2012.